MIFTAHUL MUHAEMEN

Software Engineering Portfolio • Fullstack Developer (Go & React)

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FLAGSHIP PRODUCTION FULLSTACK PROJECT

Point of Sale & Inventory System (Koperasi KHMB)

Oct 2026

Tech Stack: Go, Echo, React, TypeScript, Vite, PostgreSQL, Tailwind CSS, Docker, IndexedDB, WebSerial ESC/POS

- **Offline-First Architecture & State Sync:** Built cashier client with IndexedDB local persistence. Local transactions process instantly without network roundtrips; background workers flush queued sales batches with idempotency keys, eliminating duplicate deductions during network drops.
- **High-Performance Go Backend:** Clean architecture (Handlers, Services, Repositories) powered by Go & Echo. Implemented PostgreSQL single-pass Common Table Expressions (CTEs) for daily stock pre-aggregation, slashing query latency on historical reports by over 70%.
- **Hardware Integration & Peripherals:** Global keystroke-interception heuristics differentiate human keyboard typing from high-speed USB/Bluetooth barcode scanner HID bursts. Implemented direct ESC/POS thermal printing over WebSerial and WebUSB with zero UI lag.
- **Role-Based Access Control (RBAC):** Strict boundary enforcement isolating cashier terminal (/kasir) from back-office management (/barang, /stok, /laporan). Wholesale cost margins and audit trails are inaccessible to cashier tokens.
- **Testing & Observability:** Table-driven Go unit tests, Vitest frontend hook specs, Playwright E2E validation, k6 burst load testing, and Sentry error tracking with Go structured logging.

RELEVANT FULLSTACK & DISTRIBUTED ENGINEERING EXPERIENCE

Sanak Keluarga

Dec 2025

Tech Stack: React, TypeScript, Vite, Tailwind CSS, Monaco Editor, IndexedDB, Cloudflare Pages

- Interactive familial graph visualizer and schema editor deployed on Cloudflare edge.
- Client-side graph parsing, Monaco JSON editor integration, and IndexedDB local caching.

PT Arkan Jaya Nusantara

Jan 2021

Tech Stack: Go, TypeScript, React, MongoDB, Docker, GitHub Actions, Web Bluetooth

- Architected high-throughput Go telemetry ingestion services for IoT agricultural sensors.
- Built real-time React telemetry dashboard and automated multi-arch Docker image builds.

VERIFIED CLOUD & ENGINEERING CREDENTIALS

AWS Certified Cloud Practitioner

Amazon Web Services • 2024

Cloud infrastructure design, VPC networking, IAM security policies, and AWS managed container services (ECS/EKS).

Associate Data Engineer

DataCamp • 2024

Production SQL, relational schema normalization, ETL pipeline orchestration, and PostgreSQL performance profiling.